

1. Which n-gram language model can be implemented using a first-order Markov chain?

a. Unigram **b. Bigram** c. Trigram d. Cannot be implemented

2. From a CT scan image of a cancer patient, you need to differentiate the cancerous cells from the non-cancerous cells. Which Computer Vision algorithm will you use?

a. Image classification b. Semantic segmentation c. Object detection d. Instance segmentation

3. A pointer-generator has four words in the vocabulary: v_0, v_1, v_2, v_3 . Say you want to compress a given document with four words: v_1, v_2, v_3, v_4 . For the first decoding step, p_{gen} is 0.55, attention distribution is (0.4, 0.3, 0.2, 0.1), and the output distribution of the decoder is (0.2, 0.2, 0.2, 0.4). Which word is most likely to be produced?

a. v_1 b. v_2 c. v_3 d. v_4

4. In a Transformer network, the embedding size is 256 and Q, K, V size is 64. How many parameters are in the encoder-decoder attention/cross-attention layer of a decoder block? (You do not need to take bias terms into account)

a. 16384 **b. 65536** c. 49152 d. 196608

5. A neural network is trained with dropout 0.3. During test time the vector $[0.1, 0.2, 0.3]^T$ will have which of the following outputs after passing through the dropout layer?

a. $[0.03, 0.06, 0.09]^T$ b. $[0.07, 0.14, 0.21]^T$ c. $[0.33, 0.67, 1.00]^T$ d. $[0.14, 0.29, 0.43]^T$

6. How many regressions are there in the objectives of human pose estimation?

a. 14 b. 13 c. 11 d. 9

7. An object detection problem has five types of objects. You are using label smoothing 0.1. If your object classifier assigns 0.9 probability to the right class and equal probability to all the rest, what will the cross-entropy loss be?

a. 0.46 b. 0.20 c. 0.21 **d. 0.49**

8. You want to reduce the parameters of a 7×7 convolution filter without changing its receptive field. How many parameters can you reduce at most?

a. 46 b. 43 c. 40 d. 37

9. Which one of the following is not used to combat the vanishing gradient problem?

a. **Gradient clipping** b. Skip connection c. Use of ReLU d. Initialization of identity matrix

10. You want to reduce the number of computations in a convolutional neural network for image classification. Which one of the following will be the most effective?

a. **1D convolution** b. **1*1 convolution** c. **Max-pooling** d. **Auxiliary loss**

11. You are training a CNN with batch size 2. At one stage you have the following outputs. $x_1 = [0.1 \ 0.3 \ 0.5 \ 0.7]^T$ and $x_2 = [0.3 \ 0.5 \ 0.7 \ 0.9]^T$. What will be the output after layer normalization? Assume gamma and beta have no effect on the output.

a. Layer normalization is not applicable to CNNs

b. **$x_1 = [-1.34 \ -0.45 \ 0.45 \ 1.34]^T$, $x_2 = [-1.34 \ -0.45 \ 0.45 \ 1.34]^T$**

c. $x_1 = [-1 \ -1 \ -1 \ -1]^T$, $x_2 = [1, 1, 1, 1]^T$

d. None of the above

12. What will be the output dimension of a transpose convolution operation with filter size 3×3 and stride 1 on an input feature map of size 5×5 ?

a. 5×5 b. 6×6 c. **7×7** d. 8×8